

Homework 4

ASEN 5090 Introduction to GNSS

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Problem 1: Starting Point

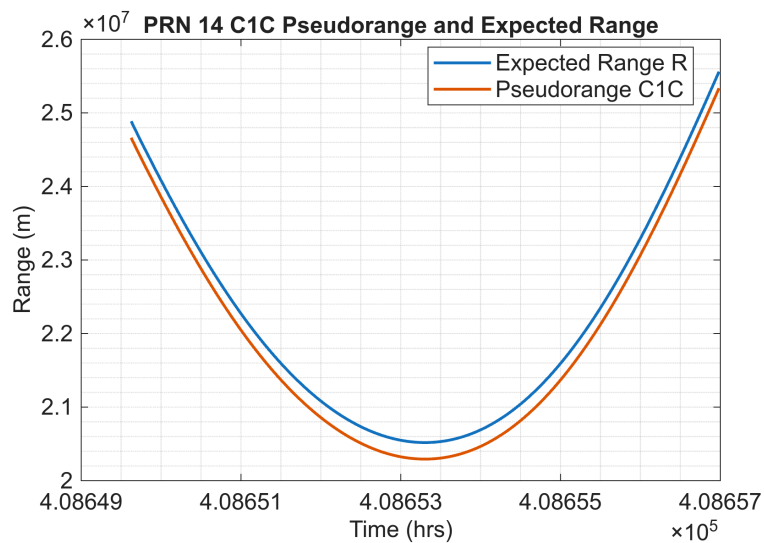


Figure 1: Expected Range and C1C Pseudorange

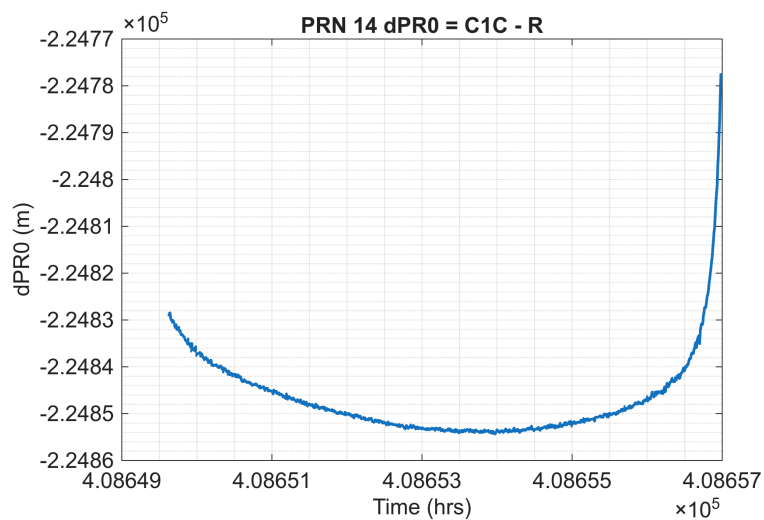


Figure 2: dPR0 Results

dPR0 first value: -224829.2307 m

dPR0 last value: -224777.3128 m

The results should look very similar to what was plotted for Homework 3 which is noticeable in Figure 2. We assume we would get difference around 10 km since expected range is calculated with orbital parameters and C1C is a signal sent that needs corrections in processing, a large part of this error is from the raw pseudoranges not accounting for clock bias.

Problem 2: Clock Error

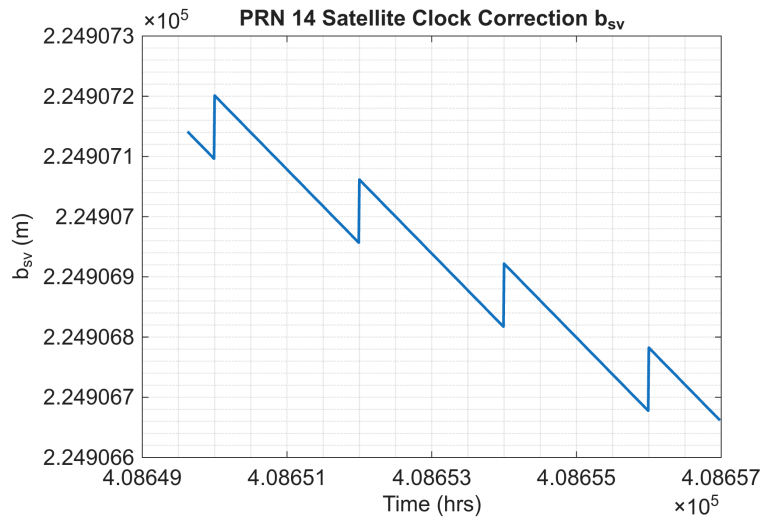


Figure 3: Satellite Clock Correction

Taking a look at Figure 3, we expect to notice a few things. The satellite clock corrections are representing clock offsets that should decrease over time, we can also notice the clock corrections occurring that is correcting GPS time to UTC(USNO) approximately every 2 hours when new broadcast ephemeris is being sent.

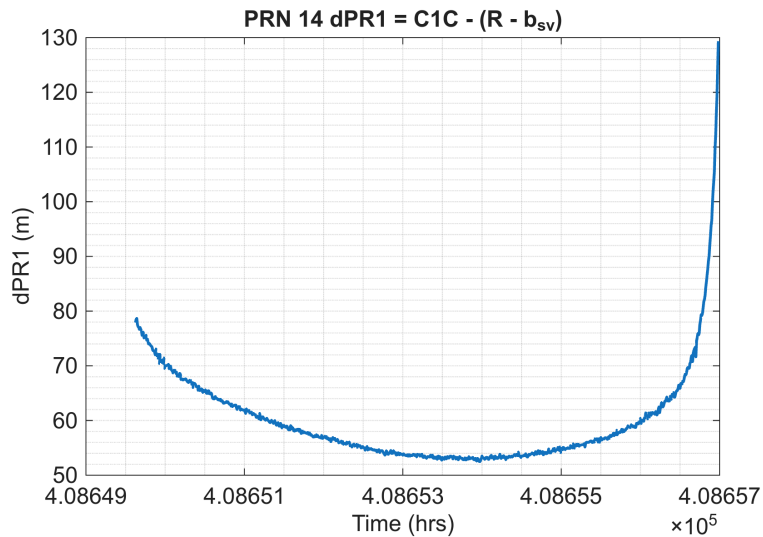


Figure 4: dPR1 Results

dPR1 first value: 77.9106 m
dPR1 last value: 129.3488 m

Problem 3: Relativity

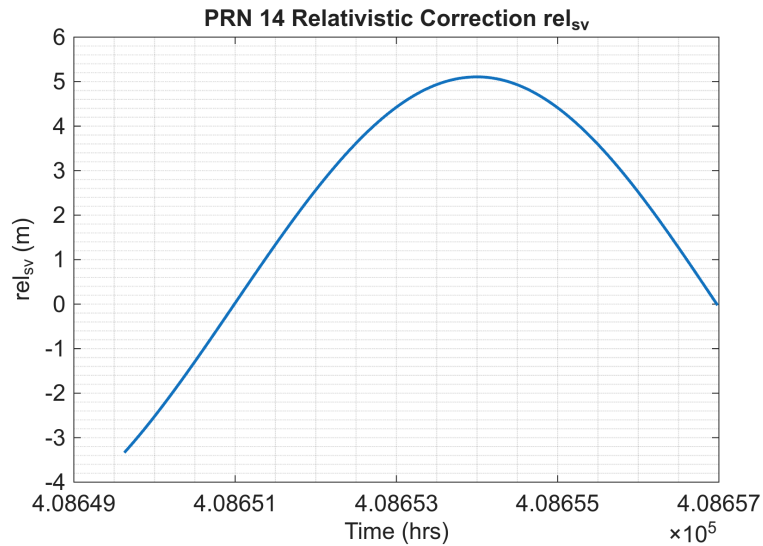


Figure 5: Relativistic Error

In our equation for the relativistic correction it is dependent on $\sin(E)$ where E is the eccentric anomaly that tells us where we are in our elliptical orbit. Since we have a $\sin(E)$ in our equation and that's the parameter that is not a constant means the graph with time should look like a sine graph.

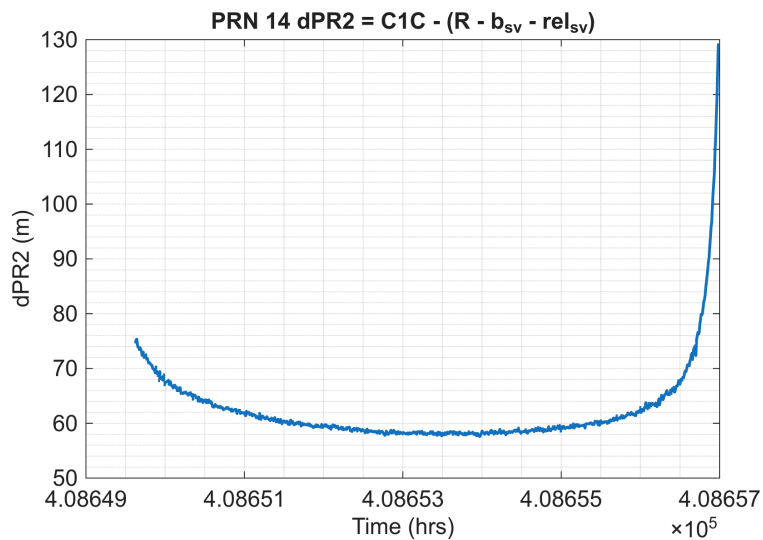


Figure 6: dPR2 Results

dPR2 first value: 74.5726 m

dPR2 last value: 129.3236 m

Problem 4: Troposphere

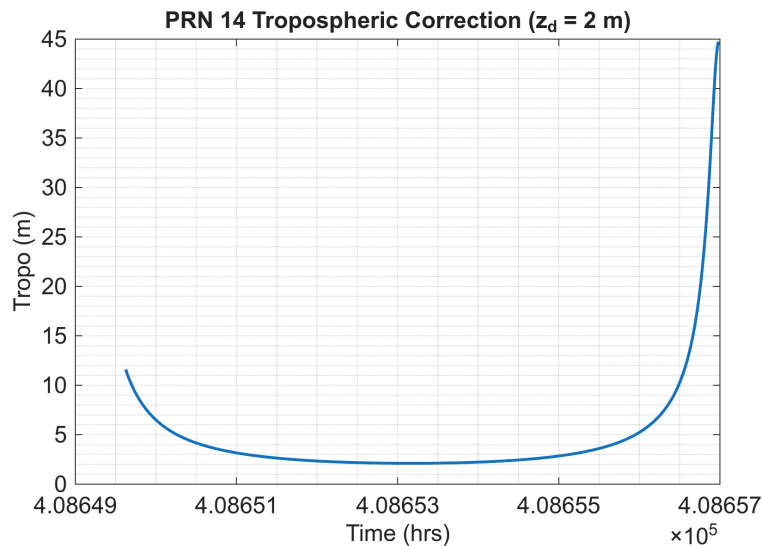


Figure 7: Tropospheric Corrections

When plotting Tropospheric Corrections, we would expect Figure 7 to exhibit characteristics that we would see from tropospheric effects. We would typically see large errors near the start and end of a satellite pass due to the tropospheric delay being proportional to the propagation distance which has a dependence on the incidence angle.

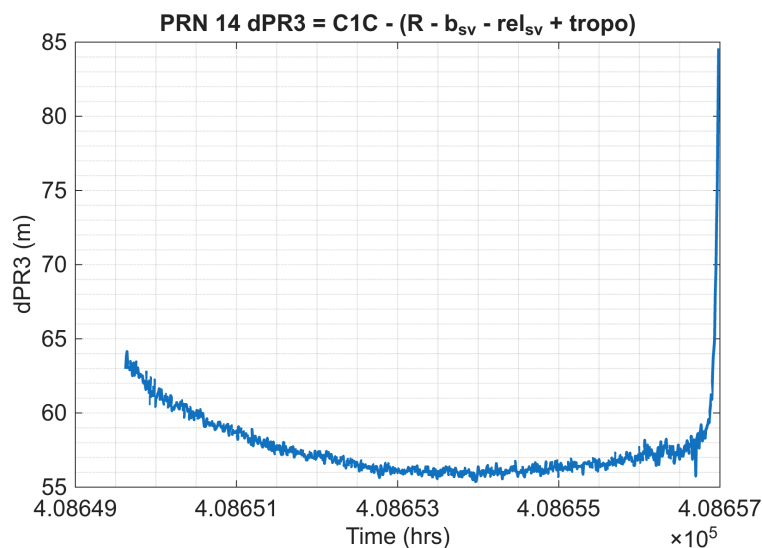


Figure 8: dPR3 Results

dPR3 first value: 62.9696 m
dPR3 last value: 84.5804 m

Problem 5: Ionosphere

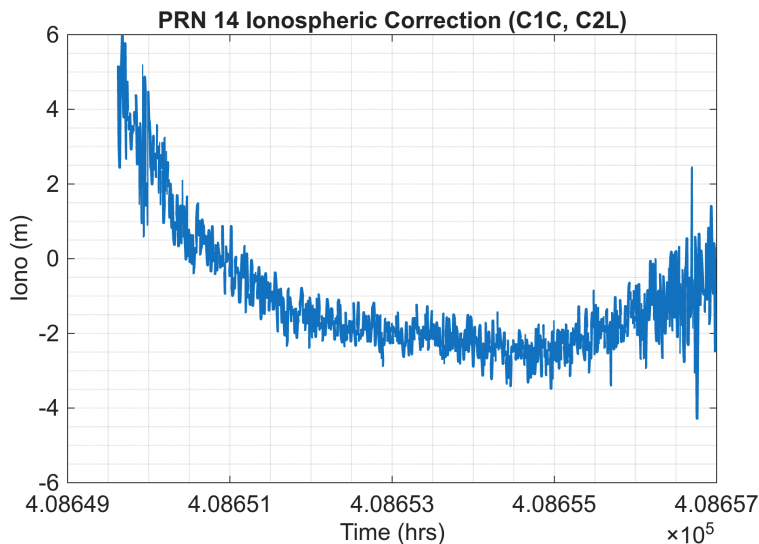


Figure 9: Ionospheric Correction (C1C, C2L)

We would expect a varying curve since ionosphere is different in regions. We would expect it to look noisy in the signal since we are mixing frequencies and ranges. There should be some positive delay of a handful of meters, where it is larger at the start and end of a pass at lower elevation angles. From dual frequencies being used we would naturally also see a relatively noisy result.

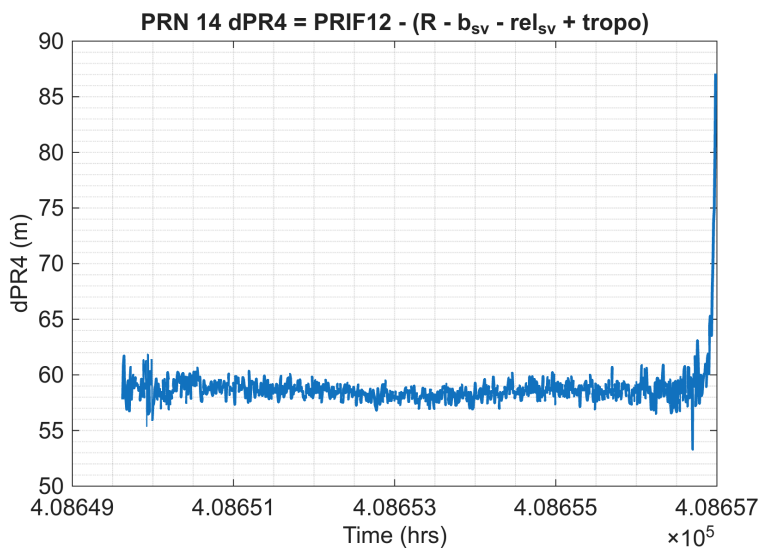


Figure 10: dPR4 Results

dPR4 first value: 57.7976 m

dPR4 last value: 87.0845 m

Problem 6: Discussion

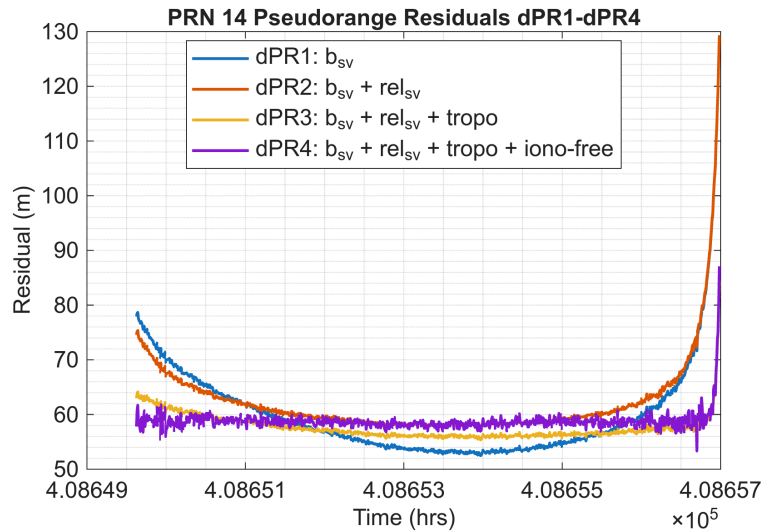


Figure 11: All dPR Results

Examining the results in Figure 11, we notice the cumulative effect of the corrections on the residuals. Our first correction, dPR1, immediately reduces the residuals by several orders of magnitude, to only tens of meters by accounting for clock bias effects. Accounting for the relativistic error improves the residuals in the middle of the satellite pass and slightly improves the start of the pass by 10s of meters. Incorporating the tropospheric correction significantly improves the start and end of the pass by 10s of meters as well. Finally, accounting for ionospheric effects makes the residuals much less time dependent, so they remain nearly constant over the pass by a couple meters; it should be noted that since this correction combines measurements on dual frequencies for an ionospheric free model of pseudorange, it also amplifies the measurement noise.

Problem 7: Multipath

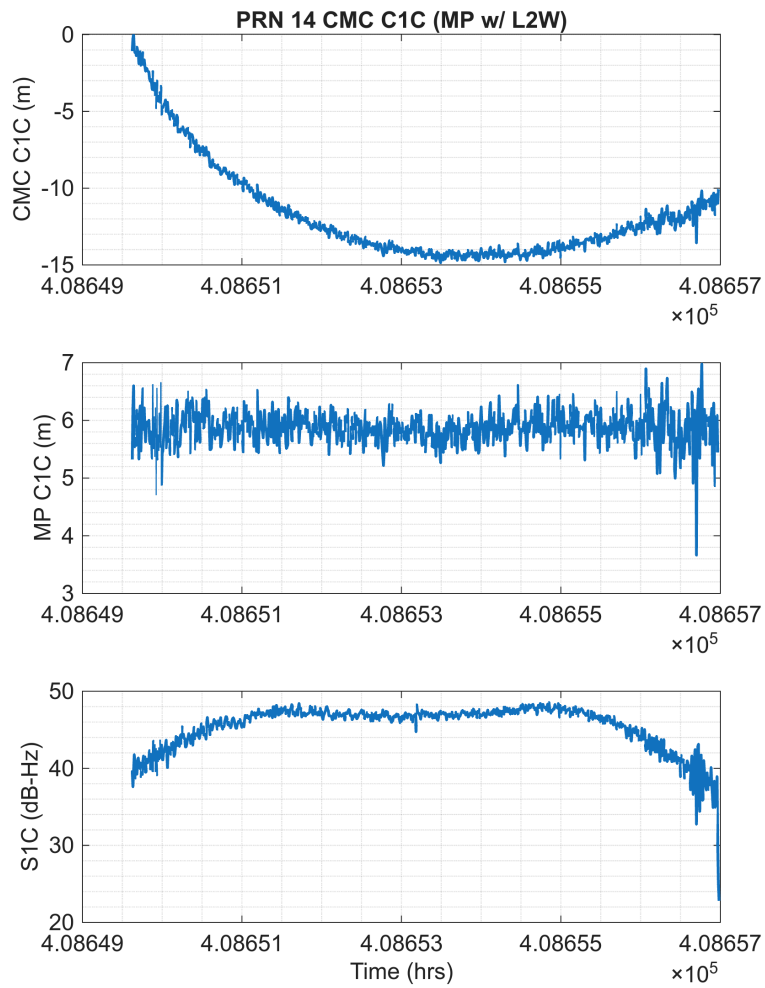


Figure 12: CMC1 Multipathing

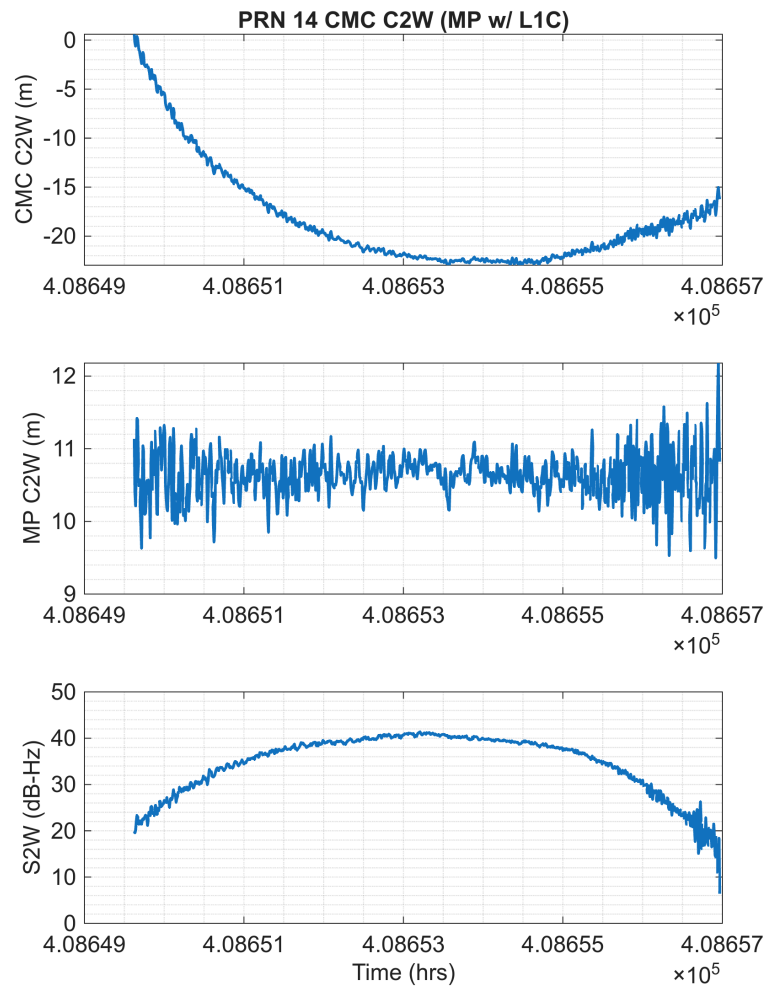


Figure 13: C2W Multipathing

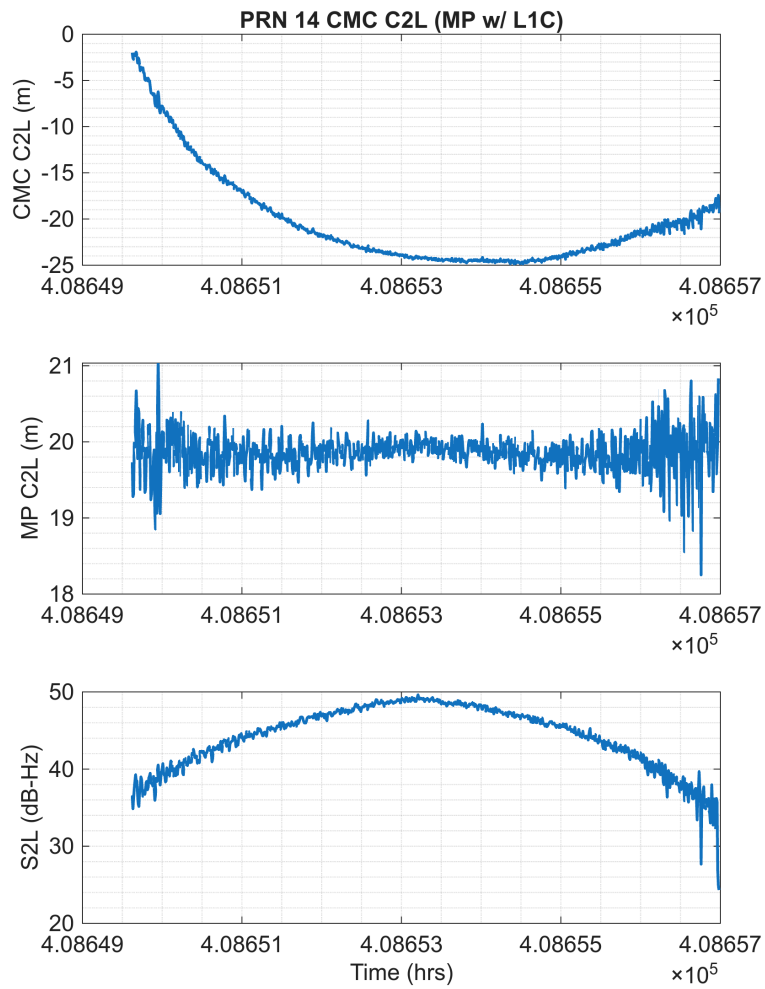


Figure 14: C2L Multipathing

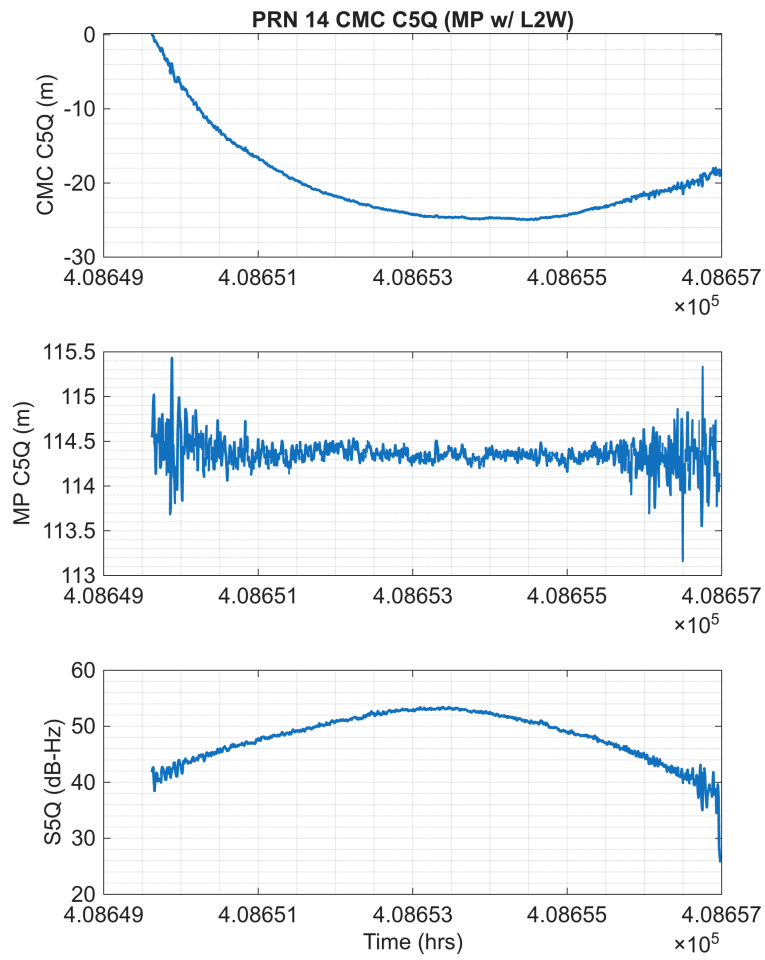


Figure 15: C5Q Multipathing

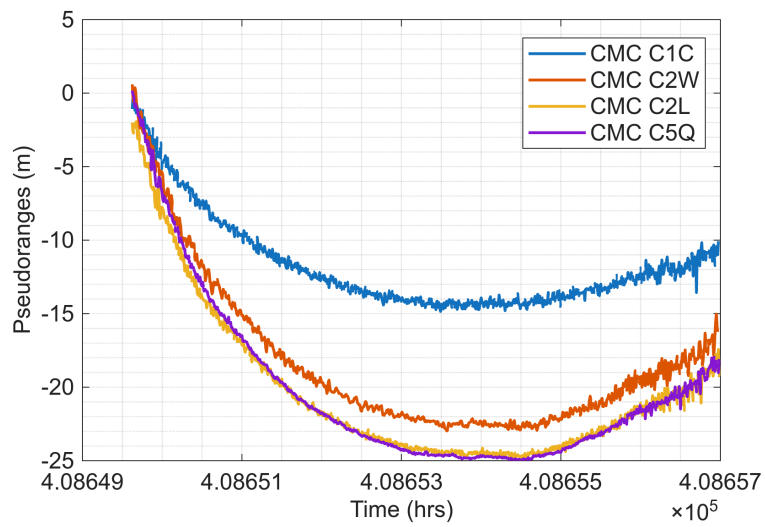


Figure 16: All Multipath Pseudoranges

Looking at results starting with Figure 12 we can see a lot of noise in C1C once we add multipathing which makes it not a good signal for very crowded cities. The noise ratio graph for S1C vs time also tells us we should not use this CA code for positioning in big cities because it's very noisy.

Then from Figure 13 in the MP C2W graph we see a lot of noise at the edges of the dataset and smaller amount in the center so it wouldn't be the best positioning item but better than C1C. S2W is much less noisy than S1C with time so it's better for positioning but still not the best.

From Figure 14 for C2L we see a similar trend to Figure 13 but we see even more noise reduction when looking at the average signal and not the edges. S2L also is almost smooth so it would be good at positioning.

From Figure 15 we can see the edges are a bit noisy but the average signal is almost smooth so this means that C5Q is the best one for multi path defense when positioning. S5Q is the most smooth graph for the SNR vs time so this is another reason why it would be the best one for positioning.

Looking at Figure 16 where all the pseudoranges are on one plot we see CMC C5Q and CMC C2L almost overlapping and C1C the furthest from the others which is evidence that C1C is the worse for positioning when multipath is happening and C5Q is the best. This makes sense because C1C is using CA code and L1 thus it's the oldest one out of the group we are analyzing and it has a broadening peak when comparing to other like C5Q. C1C is also the first civilian gps signal thus it must be more crowded so noisier.